Market Thickness vs. Capital Depth: An Agent-Based Analysis of Agricultural Contract Markets Under Liquidity Stress

AgriMarket ABM Project
Hugging Face Spaces
https://huggingface.co/spaces/rdjarbeng/agrimarket-abm

***Abstract*—We investigate whether market thickness (number of active participants) or capital depth (total investable capital) is the dominant driver of agricultural contract market outcomes. Using AgriMarket ABM, an agent-based model implemented in Mesa 3.5 featuring heterogeneous investor strategies, asymmetric information tiers, crop growth dynamics, and endogenous bank lending, we conduct a 4×3 factorial experiment across market regimes and information levels with $N = 600$ Monte Carlo simulations. Our key finding is that reducing investor count from 20 to 5 while tripling per-investor capital produces a 70% reduction in bidding competition (Cohen's $d = -9.3$), a 190% increase in zero-competition trades ($d = +2.9$), and a 46% increase in top-buyer market concentration ($d = +3.0$), while prices and total trade volume remain statistically unchanged. Holdings inequality (Gini) paradoxically decreases by 42% ($d = -8.3$) as the few remaining investors are forced to diversify. These results suggest that policies promoting market participation breadth may be more effective than capital injection programs in improving agricultural market efficiency.**

***Index Terms*—agent-based modeling, agricultural markets, market microstructure, liquidity, information asymmetry, Monte Carlo simulation**

I. Introduction

Agricultural contract markets—where investors purchase forward contracts on crop yields—play an increasingly important role in developing economies by providing farmers with capital before harvest and shifting production risk to better-capitalized agents [1]. The efficiency of these markets depends critically on two factors: the *number* of active participants (market thickness) and the *capital* those participants bring (market depth).

A central question for agricultural policy is whether capital subsidies or participation-broadening programs better serve market efficiency. Standard economic theory suggests that thin markets with deep-pocketed participants should still achieve efficient price discovery through competitive bidding. However, this assumes homogeneous agents and complete information—assumptions that rarely hold in agricultural settings.

We address this question using an agent-based model (ABM) that captures the heterogeneity, bounded rationality, and information asymmetry characteristic of real agricultural markets. Our model features:

- Five investor strategy types with distinct bidding behaviors
- Five information tiers from blind (price-only) to precision (real-time crop telemetry)
- Endogenous weather, crop growth (GDD-logistic), and disease dynamics
- Bank lending with LTV-based credit assessment
- Ornstein-Uhlenbeck commodity spot price processes

We design a 2×2 factorial experiment crossing market thickness (5 vs. 20 investors) with capital depth (normal vs. elevated), yielding four scenarios tested across three information tiers with 50 Monte Carlo replications each ($N = 600$ total simulations). We measure 14 market microstructure indicators including bidding competition, market concentration, price discovery quality, and capital efficiency.

II. Related Work

Agent-based models of agricultural markets build on foundational work in computational economics [2]. Early agricultural ABMs focused on land-use dynamics [3] and diffusion of farming practices [4]. More recent work models agricultural supply chains [5], commodity speculation [6], and the impact of information technology on smallholder markets [7].

Market microstructure theory provides the analytical framework for our liquidity metrics. Kyle's [8] lambda model establishes that market depth determines price impact per unit of trade. Glosten and Milgrom [9] show that bid-ask spreads arise from information asymmetry between informed and uninformed traders—directly analogous to our information tier mechanism.

The interaction between market thickness and efficiency has been studied in labor markets [10] and financial markets [11], but applications to agricultural contract markets remain sparse. Our contribution is an ABM that jointly models crop dynamics, heterogeneous investors, and banking within an agricultural market microstructure framework.

III. Model Description

A. Overview

AgriMarket ABM is implemented in Mesa 3.5 [12] with manual staged activation. One simulation step represents one week. The model comprises five agent types: *Plot* (farmland with crop growth state), *Farmer* (owns plots, offers contracts), *Investor* (buys contracts, trades on secondary market), *Bank* (provides loans), and *PriceManager* (dynamic ask pricing).

B. Crop Growth Model

Crop development follows a Growing Degree Day (GDD) logistic model:

$$B(t) = \frac{B_{\max}}{1 + e^{-k(G(t) - G_{\text{mid}})}} \cdot \prod_i (1 - s_i) \tag{1}$$

where $B(t)$ is biomass at time t, $G(t)$ is accumulated GDD, k is the growth rate, G_{mid} is the inflection GDD, and s_i are stress multipliers for water, temperature, disease, and flood damage.

Weekly GDD accumulation is $\Delta G = \max(0, T - T_{\text{base}}) \times 7$, where T is the weekly mean temperature. Soil moisture follows:

$$\text{SM}_{t+1} = \min(\text{FC}, \max(0, \text{SM}_t + \frac{P + I - \text{ET} - R}{100})) \tag{2}$$

where FC is field capacity, P is precipitation, I is irrigation, ET is evapotranspiration, and R is runoff.

C. Weather Engine

Temperature follows a seasonal sinusoidal model with Gaussian noise:

$$T(w) = \bar{T} + A_T \sin\left(\frac{2\pi(w - \phi)}{52}\right) + \mathcal{N}(0, \sigma_T) \tag{3}$$

Rainfall is gamma-distributed with seasonal modulation. Climate modes (normal, El Niño, La Niña) shift the mean temperature and rainfall scale parameters.

D. Spot Price Engine

Commodity prices follow an Ornstein-Uhlenbeck process:

$$dP = \theta(\mu - P)dt + \sigma dW \tag{4}$$

where $\theta = 0.1$ is the mean-reversion rate, μ is the long-run equilibrium price (crop-specific), and σ is scaled from the crop's price volatility parameter.

E. Investor Decision Model

Investors compute a contract valuation V and bid if $V >$ ask:

$$V = P_{\text{spot}} \cdot Y_{\text{exp}} \cdot (1 - r_{\text{weather}}) \cdot 0.25 \cdot \alpha_{\text{tier}} \cdot \beta_{\text{strategy}} \tag{5}$$

where P_{spot} is the commodity spot price, Y_{exp} is expected yield, r_{weather} is perceived weather risk, the 0.25 factor is the per-plot hectare fraction, α_{tier} is the information tier adjustment, and β_{strategy} is the bid aggressiveness parameter.

Information tiers modify valuation as follows:

- **Blind**: No adjustment (uses default yield/price estimates)
- **Local**: Anchors to own trade history ($0.6V + 0.4 \cdot \bar{P}_{\text{own}}$)
- **Market**: Blends with public price index ($0.5V + 0.5 \cdot P_{\text{index}}$)
- **Full**: Market + competitor count adjustment ($V \cdot (1 + 0.05 n_{\text{bids}})$)
- **Precision**: Full + disease, growth fraction, irrigation data

F. Banking Model

Banks evaluate loan applications using a loan-to-value (LTV) framework:

$$\text{LTV} = \frac{\text{amount}}{\max(C_{\text{collateral}}, C_{\text{cash}} + C_{\text{collateral}})} \tag{6}$$

Approval depends on credit score, LTV ratio, and collateral coverage, weighted $0.4 : 0.3 : 0.3$. Three bank behavior modes (conservative, moderate, aggressive) set different LTV thresholds and rate premiums.

IV. Experimental Design

A. Scenarios

We construct a 2×2 factorial design crossing market thickness with capital depth (Table I). All scenarios use 2 farmers with 16 plots each (32 total), 2 banks, tropical climate, and 30-week seasons $\times$ 2 seasons.

TABLE I
Experimental Scenarios

Scenario	Investors	Capital	Spec. %
Baseline	20	1×	25%
Thin Liquidity	5	1×	20%
High Demand	20	2×	45%
Thin + High Demand	5	3×	60%

Capital multipliers scale all investor starting capital uniformly. Strategy mixes shift toward speculators in high-demand scenarios to model aggressive market conditions.

B. Information Tier Treatment

Each scenario is tested across three information tiers: blind, market, and precision. This yields a $4 \times 3 = 12$ cell design.

C. Monte Carlo Protocol

Each cell is replicated 50 times with seeds $s = r + h(l) \bmod 10000 + h(c) \bmod 5000$, where r is the run index, l is the information level, and c is the scenario name. This ensures matched weather sequences within tiers while varying across scenarios. Total simulations: $N = 4 \times 3 \times 50 = 600$.

D. Metrics

We compute 14 microstructure metrics per simulation. Effect sizes between each scenario and baseline are quantified using Cohen's d:

$$d = \frac{\bar{x}_{\text{scenario}} - \bar{x}_{\text{baseline}}}{\sqrt{(s^2_{\text{scenario}} + s^2_{\text{baseline}})/2}} \tag{7}$$

We classify $|d| > 0.5$ as medium and $|d| > 0.8$ as large effects.

V. Results

A. Market Thickness Dominates Competition Structure

The most dramatic effects arise from reducing investor count, not from increasing capital. Table II presents key metrics.

TABLE II
KEY METRICS BY SCENARIO (MEAN ± STD, $n = 150$)

Metric	Base	Thin	High	T+H
Avg Bidders	14.6 ±1.5	3.5 ±0.3	14.9 ±1.2	4.3 ±0.4
Top Buyer %	22% ±3%	50% ±4%	17% ±3%	33% ±3%
No-Comp. %	7% ±4%	23% ±7%	8% ±3%	20% ±5%
Gini	0.70 ±0.03	0.56 ±0.04	0.54 ±0.05	0.41 ±0.04
Trades	42.4 ±8.0	41.8 ±7.4	42.7 ±8.0	42.3 ±7.4
Unsold %	31% ±14%	31% ±13%	30% ±14%	32% ±12%
Avg Price ($)	1019 ±486	990 ±466	1035 ±477	1020 ±516
ROI	15% ±6%	68% ±23%	8% ±3%	18% ±6%

Bidding competition drops 70% ($d = -9.3$) when investor count decreases from 20 to 5, regardless of capital level. This is the largest effect in the experiment by an order of magnitude. Meanwhile, doubling capital with 20 investors produces only a 2.6% increase in bidding ($d = +0.28$, nonsignificant).

B. Capital Cannot Substitute for Participation

The Thin + High Demand scenario provides a clean test: 5 investors with 3× capital have the same total capital pool as the 20-investor baseline. Yet:

- Trade count is identical: 42.3 vs 42.4 ($d = -0.007$)
- Unsold fraction is identical: 32% vs 31% ($d = +0.12$)
- Average price is identical: \$1,020 vs \$1,019 ($d = +0.004$)

The mechanism is structural: each investor evaluates one lot per step and bids on at most one. With 5 investors, the maximum bids per step is 5, creating an inescapable throughput bottleneck regardless of capital depth.

C. Paradoxical Inequality Reduction

Holdings Gini decreases monotonically with fewer investors: 0.70 (baseline) → 0.56 (thin) → 0.41 (thin+high), with $d = -8.3$ for the extreme case. This arises because with 32 plots and 5 investors, each investor accumulates ∼6–8 plots on average, producing a more uniform distribution than the 20-investor case where some investors acquire 0 plots and others acquire 4–5.

This reveals a measurement artifact: Gini measures inequality *among active participants*, not welfare across the entire economy. The 32% unsold plots represent farmer welfare losses not captured by investor-side Gini.

D. Thin Markets Reward Survivors

Thin Liquidity investors earn 68% ROI ($d = +3.2$) versus 15% in the baseline—a 4.4× multiplier. Capital velocity (trade volume / initial capital) increases from 0.13 to 0.51 ($d = +1.6$). With minimal competition, investors buy at favorable prices and deploy capital more efficiently. This "thin market premium" is analogous to illiquidity premia observed in financial markets [11].

E. Information Tier Interaction

TABLE III
AVERAGE TRADE PRICE BY INFORMATION TIER AND SCENARIO

Scenario	Blind	Market	Precision
Baseline	\$798	\$626	\$1,632
Thin + High	\$803	\$612	\$1,646

Information tier effects are remarkably stable across market regimes (Table III). Precision-tier investors consistently pay ∼2× the blind-tier price. The market thickness manipulation does not significantly interact with information tier—contrary to our hypothesis that precision information would provide greater advantage in thin markets.

F. Effect Size Summary

Table IV presents Cohen's d for the Thin + High Demand scenario versus baseline.

TABLE IV
EFFECT SIZES: THIN + HIGH DEMAND VS. BASELINE

Metric	d	% Change	Size
Avg Bidders	−9.33	−70.5%	Large
Gini Holdings	−8.29	−41.8%	Large
Top Buyer Share	+3.03	+46.1%	Large
No-Competition	+2.94	+190.0%	Large
Bid-Ask Spread	−2.59	−10.4%	Large
Loan Default Rate	+0.40	+21.6%	—
Investor ROI	+0.37	+14.4%	—
Avg Trade Price	+0.004	+0.2%	—
Num Trades	−0.007	−0.1%	—

Five metrics show large effects ($|d| > 0.8$), all related to market structure (competition, concentration, equality). Zero metrics related to aggregate outcomes (price, volume, unsold fraction) show significant effects.

VI. Discussion

A. Implications for Agricultural Policy

Our results suggest a clear policy hierarchy: **market thickness interventions dominate capital injection programs**. A market with 20 modestly-capitalized investors outperforms

one with 5 well-capitalized investors on every microstructure metric (competition, concentration, price discovery) while producing identical aggregate outcomes (volume, prices, unsold inventory).

This has direct implications for agricultural development programs:

1) Programs that increase the number of active market participants (cooperatives, mobile-enabled market access, reduced entry barriers) address the binding constraint.
2) Capital subsidies and credit programs, while valuable for other reasons, do not improve market microstructure when participation is already thin.
3) Regulatory attention should focus on concentration: in thin markets, a single buyer captures 33–50% of trades, creating monopsony conditions.

B. The Gini Paradox

The counterintuitive Gini decrease highlights a well-known limitation of inequality measures applied to small populations. With 5 investors and 32 plots, the mathematical maximum Gini (one investor owns everything) is approximately 0.8, while the minimum (equal distribution) is approximately 0.0. The observed 0.41 reflects moderate spreading among the few participants. Policy analysis should supplement investor-side Gini with farmer-side metrics (unsold fraction, revenue distribution) for a complete picture.

C. Limitations

Several limitations constrain generalizability:

- **Single-lot bidding:** Investors bid on one lot per step. Real markets allow portfolio-level bidding strategies.
- **No entry/exit:** Investor count is fixed. Dynamic entry in response to profit opportunities would moderate thin-market effects.
- **Homogeneous farmers:** All farmers use identical offering strategies. Differential pricing would add strategic complexity.
- **No spatial dimension:** Transport costs and geographic clustering are not modeled.
- **Crop sensitivity:** Results depend on crop mix. High-value crops (tomato: \$1,749/trade) produce thinner markets than staples (maize: \$319/trade) at the same investor count.

D. Future Work

Three extensions would strengthen these findings:

1) A **factory buyer agent** (demand-side) that purchases produce after harvest, creating a complete supply chain.
2) **Post-harvest losses** that interact with lot sizing decisions, adding a farmer-side adaptive mechanism.
3) **Dynamic investor entry/exit** conditioned on profitability, testing whether thin-market premia attract new entrants and self-correct.

VII. Conclusion

Using 600 Monte Carlo simulations of an agricultural contract market ABM, we demonstrate that market thickness—not capital depth—is the binding constraint on market efficiency. Reducing investor count by 75% while tripling per-investor capital collapses bidding competition ($d = -9.3$), concentrates market power ($d = +3.0$), and eliminates price discovery in 20% of trades ($d = +2.9$), while leaving aggregate prices and volume unchanged. The finding that five investors with $3\times$ capital produce identical market volume to twenty investors with $1\times$ capital establishes a structural throughput limit driven by participation count, not capital availability. These results argue for agricultural policies that prioritize market access and participation breadth over capital injection.

Reproducibility

The simulation is open-source and interactive at https://huggingface.co/spaces/rdjarbeng/agrimarket-abm. All experiments can be reproduced via the Experiment Lab mode with seed-based deterministic execution. Source code: Mesa 3.5 + Streamlit + Plotly, deployed as a Docker Space on Hugging Face.

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